

Two Pipelines

Scanned CRDP PDFs -> Markdown + LaTeX. One corpus, routed by language.

1. Inventory (free)

pdffonts / pdfimages -list / pdftotext
Sort: born-digital | clean scan | poor scan

Born-digital -> done, \$0

pdftotext -layout. No model call.
Typically 10-20% of the corpus.

Scanned -> route by language

FR / EN maths + sciences -> Pipeline A
Arabic-script subjects -> Pipeline B

PIPELINE A - French / English

Mathématiques, Physique, Chimie, Sciences de la Vie

2. Extract pages

pdfimages -png (original pixels) or
pdftoppm -r 300 -gray. Keep the PNGs.



3. Layout pass

Bounding boxes for figures and tables.
Local, free. Skips blank / index pages.



4. Primary read: Marker or MinerU

Self-hosted on GPU. PDF -> Markdown with LaTeX. Mathpix is the paid alternative.



5. Gates (free, deterministic)

KaTeX compiles every $\$...\$$ (same renderer the app uses) + Tesseract digit diff + repetition-loop check.



6. Escalate what failed

Flagged pages -> vision LLM second pass
-> human review with the page image.

100 pages: \$0.35 (self-hosted) to \$1.30 (Mathpix)

PIPELINE B - Arabic script

Arabic, Civics, History, Geography, Philosophy (AR)

2. Extract pages

pdftoppm -r 300 -gray. Keep the PNGs.



3. Layout pass

Boxes for maps, photos and tables.
Not optional here - maps ARE the content.



4. Primary read: vision LLM

Claude Opus 5 via Batch API. Handles RTL, two-column Arabic, poetry, tashkeel.



5. Gates (cheap, deterministic)

Doc AI second reader + digit diff (after normalising Arabic-Indic numerals) + NFKC / presentation-form / tatweel checks.



6. Escalate what failed

Flagged pages -> second pass at higher DPI
-> human review with the page image.

100 pages: ~ \$3.45

Identical output from both: `pages/<sha>/page-NNN.png` `text/<sha>/page-NNN.md` `figures/` `meta/<sha>/page-NNN.json`

Next phase: *split into questions and course chunks* -> *file under chapters* -> *embed* -> *pgvector*

What it costs

Per 100 pages, all-in, including the second reader and escalation.

Step	Pipeline A (FR / EN)	Pipeline B (Arabic)	Note
Page extraction	\$0	\$0	poppler, local
Layout / bounding boxes	\$0	\$0	local model or OpenCV
Primary read	\$0.05 - \$1.00	\$3.00	A: GPU minutes or Mathpix. B: Opus 5 batched
Second reader	\$0 (Tesseract)	\$0.15 (Doc AI)	Tesseract is unusable on Arabic
Gates: KaTeX, digits, Unicode	\$0	\$0	deterministic, no model
Escalation (~8 pages)	\$0.30	\$0.30	vision LLM second pass
TOTAL / 100 pages	\$0.35 - \$1.30	~ \$3.45	vs \$3.80 if everything went through the LLM
TOTAL / page	\$0.004 - \$0.013	~ \$0.035	

Across the whole corpus

Scope	Pages	Cost	Note
Pipeline A - FR/EN sciences	~10,500	\$37 - \$137	maths, physics, chemistry, life sciences, both languages
Pipeline B - Arabic subjects	~4,500	~\$155	Arabic, civics, history, geography, philosophy
TOTAL	~15,000	\$190 - \$290	against ~\$570 if every page went through the vision LLM

Assumptions to confirm on the golden set

- 5,000 input tokens + 2,000 output tokens per page for the vision LLM; Arabic prose runs lower.
- Batch API discount of 50% applied to every LLM call. Prompt caching on top of that.
- Escalation rate of 8%. This is the number most likely to be wrong - measure it, do not assume it.
- Mathpix and Google Document AI per-page rates: check current published pricing before committing.
- Self-hosted Marker / MinerU costed as GPU rental time, not per page.
- 70 / 30 split between Latin-script and Arabic-script pages. Confirm from the manifest once collection is done.

The gap between the two pipelines is the whole argument for routing: the same 100 pages cost roughly ten times more through the vision model. Run both over the 200-page golden set before committing - the deciding number is accuracy per page, not price.